Comprehensive Physics Based TCAD Model for 2D MX_2 Channel Transistors

D. Mahaveer Sathaiya[1], Terry Y.T. Hung[2], Edward Chen[2], Wen-Chia Wu[3], Aslan Wei[1], Chih-Piao Chuu[2], Sheng-Kai Su[2], Ang Sheng Chou[2], Cheng-Ting Chung[4], Chao-Hsin Chien[3], Han Wang[2], Jin Cai[4], Chung-Cheng Wu[1], Iuliana P. Radu[2], and Jeff Wu[1]

[1]TCAD, Taiwan Semiconductor Manufacturing Company, Hsinchu, Taiwan. [2]Corporate Research, Taiwan Semiconductor Manufacturing Company, Hsinchu, Taiwan. [3]Institute of Electronics, National Yang Ming Chiao Tung University, Hsinchu, Taiwan. [4]Pathfinding, Taiwan Semiconductor Manufacturing Company, Hsinchu, Taiwan. email: sathaiya@tsmc.com

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***Abstract*—**For the first time, a comprehensive TCAD model is developed to unambiguously extract key device parameters: contact resistance (R_c), channel mobility (μ_{CH}), Schottky barrier height (SBH), & D_{it} from experimental data on back-gate (BG) transistors with MX_2 channel. The model is tested and validated against three different data sets with different contact metal, quality of channel, contact, and interfaces. Using model's output, we analyze the accuracy of R_c and μ_{CH} extracted by the TLM method and provide guidance on the limits of its applicability. Finally, the model is used to project contact requirements (SBH ~0eV, high doping density > 2e13 cm^{-2}) for performant, scaled transistors with 2D material channel in stacked nanosheet configuration.

I. Introduction

Transistors with two-dimensional (2D) materials have been proposed as the enabler for extreme dimensional scaling to device pitches below 40nm [1]. Recent experimental advances in contact to 2D materials have increased interest in these materials [2, 3,4,5]. The low-reported contact resistance of 130 Ω-um [2], while larger than that for contacts to Si, would result in a transfer length of less than 5nm, if a simple transmission line measurement (TLM) is used for extraction. This short transfer length has not been validated experimentally and accuracy and applicability limits of the TLM to these very thin channels is not fully understood.

Extremely scaled contacts to 2D materials have been reported with contact metals which form Schottky barriers [6]. The associated device TCAD models described well experiments with high Fermi level pinning in the contacts. There, the impact of the metals which form a chemical bond with the 2D material was captured through the full depletion of carriers under the metal contact [7]. With the discovery that soft (semi)metals can form low resistance contacts, the need for predictive device models describing them has arisen.

In this paper, we develop a physics based TCAD model which captures well the behavior of contact both with and without the SBH. The contact is described through a vdW gap between the semi-metal and 2D which is treated as a tunneling gap in the model. This is justified experimentally by the low adhesion of semimetals [8]. The presence of the vdW gap can be predicted from the first principles (Fig. 2).

II. Model

Comprehensive TCAD model used in this work, for extracting key electrical parameters from the BG-MoS_2 device shown in Fig. 1, includes: i) DFT band structure calibrated charge (Multivalley DoS) model, ii) WKB approximated Schottky Barrier Height (SBH) tunneling (Nonlocal tunneling) via vdW gap for R_c, iii) Source Drain (SD) metal resistance, iv) energy dependent trap charge (D_{it}) for capturing the sub-threshold swing, v) Fixed charge (Q_F) for threshold voltage, vi) Phonon mobility (μ_{PH}) model, & (vii) Remote coulomb scattering mobility (μ_{RCS}) model for capturing the D_{it} & Q_F induced coulomb scattering (Table 1). DFT calculated SD metal adhesion strength on MoS_2 & the corresponding interlayer distance (i.e., vdW gap) for few metals are shown in Fig. 2, which are used as an initial model guess for the model calibration to data. Fig. 3 shows key physics insights that are captured by the SBH model used in this work. Simulated conduction band energy with the SBH at V_{BG}=0.5V (~$V_{FlatBand}$) and e^- tunneling distance at V_{BG}=22V are shown in Fig. 3(a). Also shown, in Fig. 3, are the SBH model predicted current density distribution, the current spreading, and the corresponding transfer length (L_T) which is approximately the length from the contact edge to a point where the current density is 10% of current density at the channel center.

III. Results and Discussions

A. Model Validation

This model is validated against three different data set from Bismuth (Bi) & Antimony (Sb) contacted 2D back-gate (BG) MOSFET on 20nm SiO_2/Si and 100nm SiN/Si substrates with varying SBH, vdW gap, Q_F, D_{it}, & μ_{CH}. First, the model is calibrated to sub-threshold I_d-V_{BG} data of L_{CH} 0.5 to 5um to determine the Q_F, D_{it} & SBH -0.05eV (Fig. 4(b)). Then the L_{CH} 5um (which is a μ_{CH} dominated device) and L_{CH} 0.5um (which is both μ_{CH} & R_c dominated device) are used to calibrate μ_{CH} & R_c (i.e., vdW gap). TLM extracted mobility & DFT predicted vdW gap are used as calibration initial guesses for μ_{CH} & R_c respectively. μ_{RCS} screening parameters are calibrated (c_exp=0.55 & Nc_exp1=Nc_exp2=2.5) to match the I_d-V_{BG} data near device turn-on & μ_{PH} = 78 cm^2/Vs is calibrated to match the I_d-V_{BG} data at high V_{BG}. This Bi contacted BG-MoS_2 data yield a higher vdW gap of 5.2A,

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which is attributed to a weaker adhesion strength of Bi to MoS_2 for this data, than the DFT predicted 3.2A.

The model thus calibrated is tested on another batch Bi contacted L_{CH} 100nm BG-MoS_2 data on a 100nm SiN (Fig. 5). vdW gap of 3A & SBH -0.05eV. To test the model with a SBH, we used Sb contacted BG-WS_2 data. As shown in Fig. 6, the model well captures both the thermal and Schottky branch of the sub-threshold I_d-V_{BG} data with SBH 0.195eV. These data calibrated SBH of -0.05eV for Bi & 0.195eV for Sb are consistent with DFT predicted SBH reported in [3]. Thus, the comprehensive physics-based model is shown to capture the subtle changes in the device characteristics, like i) I_d-V_{BG} cross over (b/w L_{CH} 1um & 0.5um) in Fig. 4(a) & ii) Schottky branch sub-threshold characteristics in Fig. 6(b), all with reasonable model parameter values.

B. R_c & μ_{CH} extraction and comparison to TLM method

R_c & μ_{CH} extractions are compared between TCAD and TLM methods (Fig. 7). TLM R_c is extracted from the intercept of the R_{tot} vs. L_{CH} plot for different V_{ov} ($V_{ov} = V_{BG} - V_t$, $V_t = V_{BG}$ @ I_d = 1.8e-9nA/um) & μ_{CH} is extracted from its slope. TCAD extracted R_c, for the BG-MoS_2 data in Fig. 4 with vdW gap 5.2A, is 4.15 KΩ-um, which is ~12% lower vs. TLM value at carrier concentration (n_{2D}) 1.1e13 cm^{-2}. TCAD extracted R_c is ~0.42 KΩ-um (at n_{2D} 1.1e13 cm^{-2}) from another batch Bi contacted BG-MoS_2 data (in Fig. 5) is much lower with vdW gap of 3A. TCAD extracted μ_{CH} is close to TLM extracted value. TCAD yields different μ_{CH} for L_{CH} 5um and 0.5um due to different D_{it} (Fig. 4(b)). L_{CH} 0.5um has higher D_{it} causing lower μ_{CH} at low n_{2D} due to stronger μ_{RCS}. Experimental data μ_{CH} is well explained by TCAD μ_{CH} from first principle once Q_F + D_{it} induced μ_{RCS} added to the model (Fig. 8). First principle μ_{CH} is predicted by linearized Boltzmann transport equation (LBTE) solver [9] with DFT multi-valley EMA MoS_2 band structure, DFT phonon scattering rates from [10] & RCS with dielectric screening factor 0.18.

Next, we use the comprehensive TCAD model to generate two typical I_d-V_{BG} characteristics, one with vdW gap 3A & μ_{PH} 70 cm^2/Vs and another with vdW gap 1A (low R_c) & μ_{PH} 28 cm^2/Vs, for quantifying the errors in the TLM extractions and provide suggestions to minimize the error. Also, the TCAD generated L_{CH} 5 to 0.5um I_d-V_{BG} characteristics assumes varying D_{it} as per in Fig. 4(b) for both the cases.

For the case with vdW 3A & μ_{PH} 70 cm^2/Vs, the TLM R_c error is in the range of +41% to +115% (Fig. 9(c)), which is caused by the varying D_{it} induced μ_{RCS}. Thus, any variations in data could largely impact the TLM extracted R_c. TLM extracted μ_{CH} is ~-10% to -22% lower than TCAD (Fig. 9 (d)) and can be used as an initial guess for TCAD model calibration to data. For the case with lower R_c (vdW 1A) & low μ_{PH} 28 cm^2/Vs, TLM R_c error range get bigger to +170% to +530% (Fig. 10 (b)), which is reduced to -50% to +30% by choosing the L_{CH} data points that are closer to the horizontal line in the normalized $(R_{tot}-2*R_c)/L_{CH}$ vs. L_{CH} plot as shown in Fig. 10 (a) and Fig. 9(b). Hence, we recommend using wide range of L_{CH} splits (5um to 0.1um) on the normalized $(R_{tot}-2*R_c)/L_{CH}$ vs. L_{CH} plot to screen out the outlying L_{CH} data points for minimizing the TLM R_c error to some extent. Also, we expect the TLM R_c error to increase for higher SBH as it can introduce further uncertainty in the n_{2D} calculations from the V_{ov} ($n_{2D}=q*C_{back\text{-}insulator}*V_{ov}$). **So, for the unambiguous extraction of R_c & μ_{CH}, we recommend using the rigorous TCAD comprehensive model calibration method demonstrated in this paper**.

C. Model predictions

This data validated model is used to predict the scaled technologies $L_{contact}$ 20nm R_c vs. SBH for vdW gap 3A & 1A in Fig. 11. The stronger R_c sensitivity to SBH with 1A vdW gap is due more sensitive e^- tunneling distance to SBH. Fig. 12 shows the model predicted R_c vs. $L_{contact}$ for vdW 3A to 1A emphasizing i) the requirement of n_{2D} > 2e13 cm^{-2} beneath the contact for reducing R_c & ii) how L_T, the effective length that contributes to current, decreases with decreasing vdW gap. n_{2D} > 2e13 cm^{-2} is needed for achieving R_c < 166 Ω-um for a scaled 20nm $L_{contact}$ with SBH -0.05eV and vdW gap of 2A. Finally, the model predicted R_c of 166 & 27 Ω-um (Fig. 12 (b)) are used to estimate the number of stacked 2D-sheets of 9 & 5 required for achieving IRDS 2021 updated [11] high density (HD) Ion (for year 2034) of 55uA at Ioff 100 pA/um & V_{DD} = 0.55V respectively. Device structure used for the no. of 2D-sheets estimation is a L_G 12nm 2D-GAA with μ_{CH} 100 cm^2/Vs (w/o μ_{RCS}), W_{sheet} 20nm, SD & extension doping of 2e13 cm^{-2} & CGP 38nm.

IV. Conclusions

Demonstrated the use of comprehensive physics based TCAD model for unambiguous extraction of not only R_c & μ_{ch}, but also SBH, vdW gap & D_{it} from the experimental 2D-channel BG-MOSFET data. TCAD model is shown to well capture the subtle changes in the device characteristics for three different data set with varying SBH, vdW gap, Q_F, D_{it}, μ_{CH} & carrier concentration. TCAD generated typical I_d-V_{BG} characteristics are used to show TLM is not a good method to extract R_c. Hence, for accurate extraction of R_c & μ_{CH}, we recommend using the rigorous TCAD comprehensive model-based method presented in this work. Finally, the model predicts the required number of stacked 2D-sheets of 9 & 5 for achieving IRDS 2021 HD Ion_{2034} of 55uA for R_c of 166 & 27 Ω-um respectively.

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References

[1] Z. Ahmed et. al., *IEDM*, 22.5, 2020. [2] P-C Shen et. al., *Nature,* 593, 211, 2021. [3] A-S Chou et. al., *IEDM*, 7.2, 2021. [4] K. P. O'Brien et. al., *IEDM*, 7.1, 2021. [5] A. Kumar et. al., *IEDM*, 7.3, 2021. [6] Q. Smets et. al., *IEDM*, 23.2, 2019. [7] G. Arutchelvan et. al., *Scientific Reports,* 11, 6610, 2021. [8] M-Y Li et. al., *VLSI,* 2022. [9] Z. Stanojevic et. al., *IEEE Trans. Electron Devices*, 68, 11, 2021. [10] Z. Jin et. al., *Phys. Rev. B*, 90, 045422, 2014. [11] IRDS, 2021

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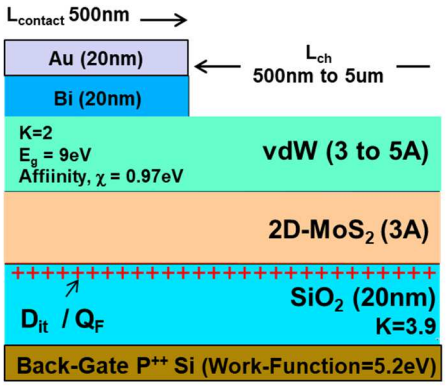

Fig. 1. BG-MoS_2 device dimensions (not to scale)

#	Models	Model parameters / Methodology
1	Charge: DoS	Q-V calibrated to DFT MoS_2 & WS_2 band structure
2	Contact Resistance (R_c): Schottky Barrier Height (SBH) & vdW gap	SBH tunneling via vdW gap (WKB approximate tunneling probability) i. SBH = ϕ_{metal} - χ_{MoS2} ii. Tunneling mass m_t~0.93m_0 (comparable to m_0 as per DFT) iii. A*= degeneracy*m_t*A_0 = 1.86 A_0 (A_0: free electron Richardson constant & A* ∝ available DOS in MoS_2) iv. Calibrate vdW gap vs. data (2A to 5A ∝ SD metal adhesion)
3	Metal resistance: Au, Bi & Sb	i. Au: ρ=2.44e-6 Ω-cm ii. Bi / Sb: ρ=3e-3 Ω-cm (Bi ρ ~ 6x of TiN, high due to semimetal)
4	Sub-threshold Swing & V_t: D_{it} & Q_F	Energy dependent D_{it} & Q_F located at MoS_2-SiO_2 interface i. Calibrate D_{it} & its energy distribution vs. IV sub-threshold slope ii. Calibrate Q_F vs. V_t
5	Mobility (μ_{CH}): $\mu_{CH}^{-1} = \mu_{PH}^{-1} + \mu_{RCS}^{-1}$	i. Calibrated μ_{PH} vs. IV data to match I_d-V_{BG} data at high V_{BG} ii. Calibrate μ_{RCS} screening to match turn-on (RCS from Dit/Q_F)

Table 1. Model parameters and its calibration methodology

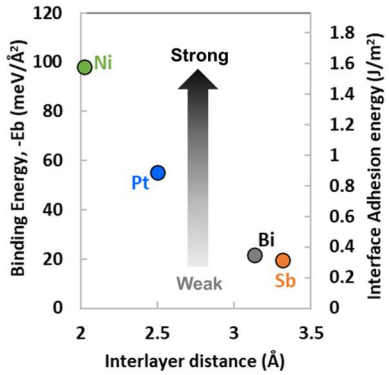

Fig. 2. DFT calculated Source Drain (SD) metal adhesion strength on MoS_2 & its Interlayer distance (i.e., van der Waals (vdW) gap) to MoS_2.

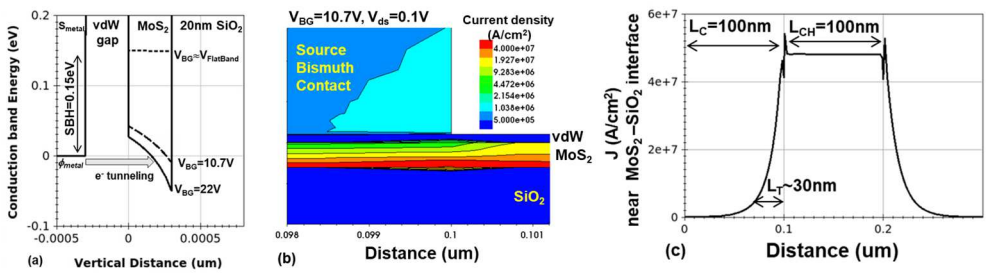

Fig. 3. Simulated SBH tunneling through a 1A vdW gap of a BG-MoS_2 with SBH 0.15eV. (a) Conduction band energy along a vertical line at the center of the Source contact for V_{BG}=0.5V (~$V_{FlatBand}$), 10.7V & 22V. (b) Current density contour in Bismuth and MoS_2 showing the current spreading under contacts. (c) Current density along a horizontal line near MoS_2-SiO_2 interface showing the current spreading near the contact edge and transfer length L_T ~30nm (i.e., the effective length which contributes to current).

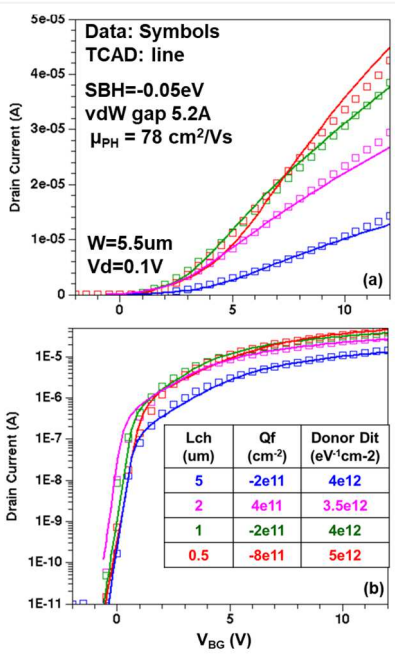

Fig. 4. Model calibration to experimental data from Bi-contacted BG-MoS_2 transistor with L_{CH} 0.5um to 5um (Fig. 1). I_d-V_{BG} on linear scale (a) and log scale (b). Model calibration is performed with vdW gap of 5.2A, phonon mobility μ_{PH} 78cm²/Vs, and Q_F + D_{it} induced μ_{RCS}. Uniform donor D_{it}, with energy distribution from E_c to 0.5eV below E_c, is used to match the sub-threshold data.

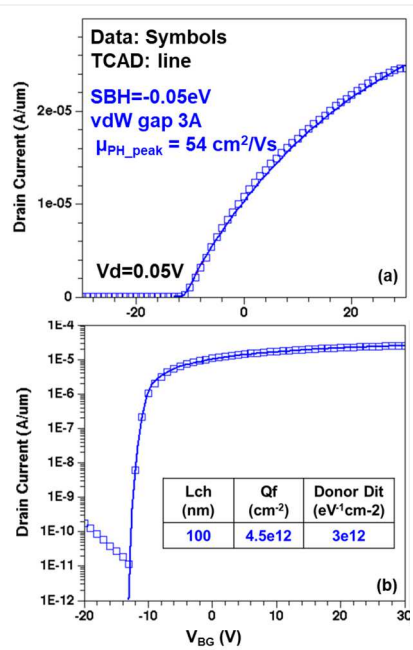

Fig. 5. Model validation to another batch of BG-MoS_2 devices with Bi contact L_{CH} 100nm; $L_{contact}$ 1um. Gate dielectric here is 100nm SiNx. I_d-V_{BG} plotted on linear scale (a) and log scale (b). Model is calibrated with vdW gap 3A, at μ_{PH_peak} 54 cm²/Vs. Uniform donor Dit along with Q_F are calibrated to the sub-threshold data.

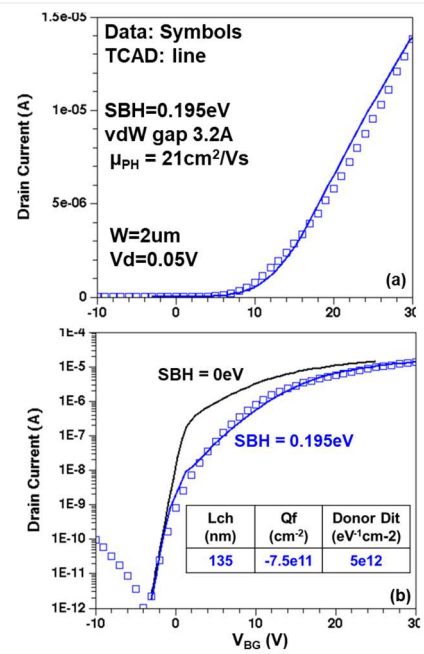

Fig. 6. Model validation to Sb-contacted BG-WS_2 transistor with L_{CH}=135nm and $L_{contact}$=1um. Gate dielectric is 100nm SiNx. (a) I_d-V_{BG} plotted on linear scale. (b) I_d-V_{BG} on log scale. The model captures both the thermal and Schottky branch of the sub-threshold data. Model is calibrated with vdW gap=3.2A, μ_{PH} 21cm²/Vs, and Q_F + Dit induced μ_{RCS}.

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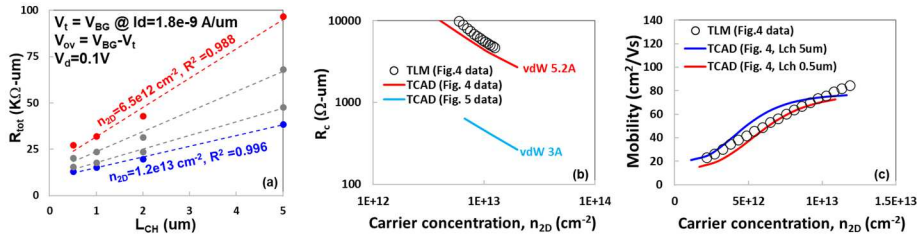


Fig. 7. Comparison of R_c and μ extracted from Fig. 4 using TLM and TCAD methods. (a) experimental R_{tot} versus L_{CH} plotted at different carrier concentration (n_{2D}). R_c is determined by the intercept of the lines while μ is inversely proportional to the slope. (b) TCAD-extracted R_c with vdW gap=5.2Å is about 12% lower than the TLM-extracted value. For comparison, TCAD-extracted R_c from experimental data in Fig. 5 with vdW gap=3 Å is much lower. (c) TCAD extracted μ_{CH} is comparable to the TLM-extracted mobility. Different μ_{CH} for L_{CH} 5um and 0.5um is induced by the different D_{it} (Fig. 4(b)). L_{CH} 0.5um has higher D_{it} causing lower μ_{CH} at low n_{2D} due to stronger remote Coulomb scattering.

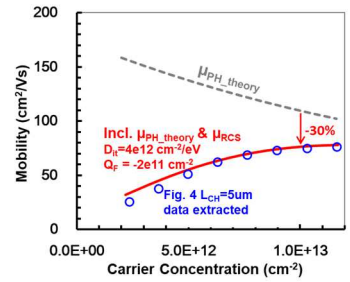


Fig. 8. Experimental data μ_{CH} is well explained by TCAD μ_{CH} from first principle once Q_F + D_{it} induced μ_{RCS} added to the model. First principle μ_{CH} is predicted by LBTE solver [9] with DFT calibrated multi-valley EMA MoS_2 band structure & DFT calculated phonon scattering rates [10]

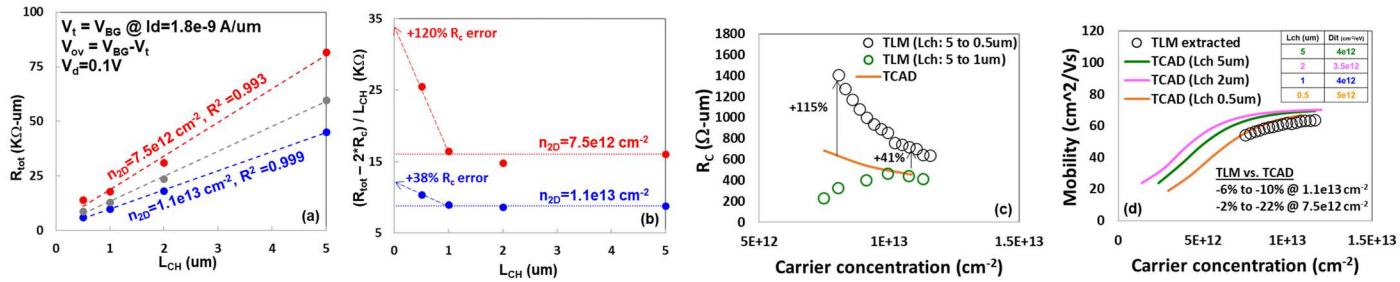


Fig. 9. TLM extraction on TCAD generated I_d-V_{BG} of a BG-MoS_2 device with same R_c (vdW 3A), but different D_{it} induced μ_{RCS} for L_{CH} 5um to 0.5um. Data thus generated is analyzed using TLM to test the precision of TLM-extracted R_c & μ_{CH}. (a) TLM R_{tot} vs. L_{CH} plot for different n_{2D} with TLM fitting R^2, (b) Normalized R_{tot} vs. L_{CH} plot showing the error in TLM extracted R_c for two different n_{2D}, (c) TLM extracted R_c vs. TCAD, also shown is the TLM extracted R_c from L_{CH} 5, 2, & 1um devices which is comparable to TCAD at high n_{2D} (green symbols). Even with restricted range of data the TLM deviates by -67% at low n_{2D} which limits the applicability of the TLM & (d) TLM extracted μ_{CH} is ~-10% to -22% vs. TCAD and can be used as an initial guess for model calibration. Also, the TLM calculated carrier concentration n_{2D} (=q*$C_{back\text{-}insulator}$*$V_{ov}$) is within +10% to the TCAD simulated (n_{2D}).

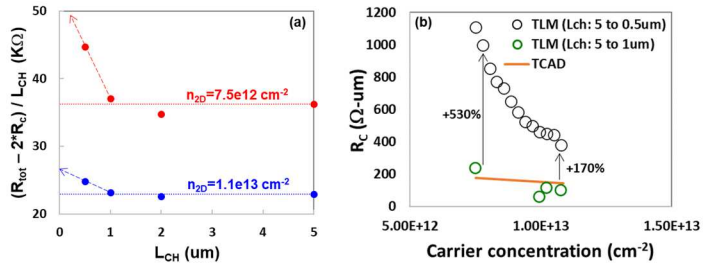


Fig. 10. TLM extraction on TCAD generated I_d-V_{BG} of a BG-MoS_2 device with low R_c (vdW 1A), low μ_{PH} ~28 cm^2/Vs & different D_{it} (Fig.9(d)) induced μ_{RCS} for L_{CH} 5um to 0.5um. For this case R_c error range from +170% to +530% vs. TCAD (b), thus TLM is not a good method to extract R_c at low resistance. However, the precision could be increased to -50% to +30% (green symbols in (b)) by choosing the L_{CH} points closer to the horizontal line in the normalized R_{tot} plot (a).

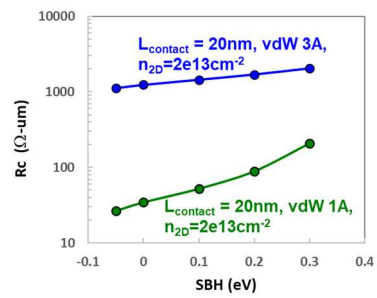
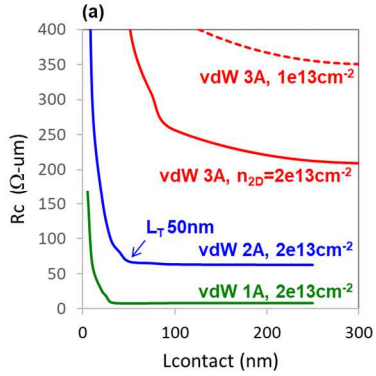


Fig. 11. Simulated R_c (excl. R_{metal}) vs. SBH of BG-MoS_2 in Fig.1. using the data validated model.



(b)

Device: BG-MoS_2 (Fig.1)						Device: L_G 12nm 2D-GAA (V_{DD} = 0.55V, μ_{CH} 100 cm^2/Vs)
vdW gap (A)	SBH (eV)	n_{2D} (cm^{-2})	L_T (nm)	$L_{contact}$ (nm)	R_c (Ω-um)	Required no. of stacked 2D-sheets to reach IRDS Ion_{2034} of 55uA at Ioff 100pA/um
3	-0.05	1e13	100	20	1750	
3	-0.05	2e13	100	20	1127	
2	-0.05	2e13	50	20	166	9
				10	304	
1	-0.05	2e13	30	20	27	5
				10	64	

Fig. 12. (a) Simulated R_c (excl. R_{metal}) versus $L_{contact}$ of BG-MoS_2 (Fig. 1) for different vdW gap and n_{2D} using the data validated SBH model. (b) Table show the transfer length (L_T) & R_c at $L_{contact}$ 20 & 10nm. Also shown is the required no. of 2D-sheets to achieve the IRDS 2021 [11] HD Ion for year 2034

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